A MECHANISTIC MULTISCALE FRAMEWORK COUPLING WITHIN-HOST RESISTANCE EVOLUTION AND BETWEEN-HOST TRANSMISSION

Felipe Schardong[1], Claudio Jose Struchiner[1], Luiz Max Carvalho[1]

School of Applied Mathematics, Getulio Vargas Foundation (FGV - EMAp)

felipe.schardong@fgv.edu.br[1], claudio.struchiner@fgv.br[1], luiz.fagundes@fgv.br[1]

Introduction

Antimicrobial resistance (AMR) is a major global health threat, responsible for an estimated 1.27 million direct deaths in 2019 and projected to cause over 39 million cumulative direct deaths between 2025 and 2050 [1, 2]. Resistance emerges within hosts through drug exposure, pathogen growth, mutation, and horizontal gene transfer [3]. Its spread at the population level depends on transmission and spatial connectivity [4, 5]. Mathematical models have addressed within-host pharmacology [6, 7] and between-host transmission [8, 9], while stochastic approaches have highlighted the role of demographic fluctuations in the establishment of rare resistant strains [10, 11]. However, these processes are often studied separately, limiting our understanding of how drug exposure and stochastic resistance evolution within hosts interact with transmission and spatial connectivity to shape the emergence and spread of AMR. Here, we develop a mechanistic multiscale framework coupling pharmacokinetics and pharmacodynamics, stochastic within-host resistance evolution, and lineage-specific between-host transmission in a two-patch metapopulation. The framework combines PBFTPK models [12, 13], pharmacodynamic selection mechanisms [14], and stochastic population dynamics [15, 16]. We use it to investigate how different patterns of treatment non-adherence [17, 18] and asymmetric mobility shape resistance emergence and spread.

Methods

An overview of the general mathematical framework is illustrated in Figure 1, linking drug dosing to population-level spatial spread.

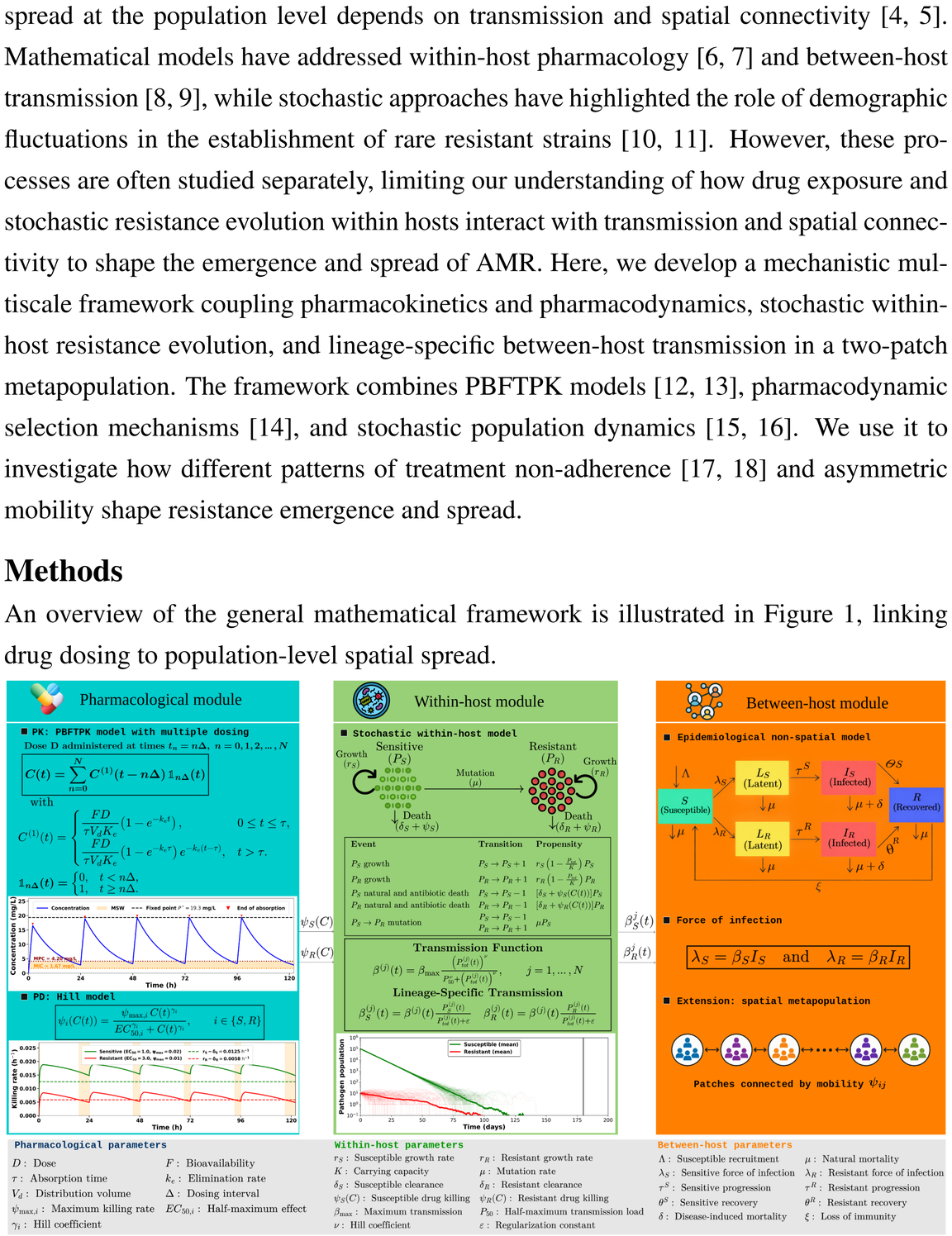


Figure 1: Schematic representation of the mechanistic multiscale mathematical framework coupling within-host resistance evolution and between-host transmission. Source: Author.

For each scenario, we define non-adherence sub-levels, summarized in Table 1, to assess how deviations from the prescribed regimen affect resistance development.

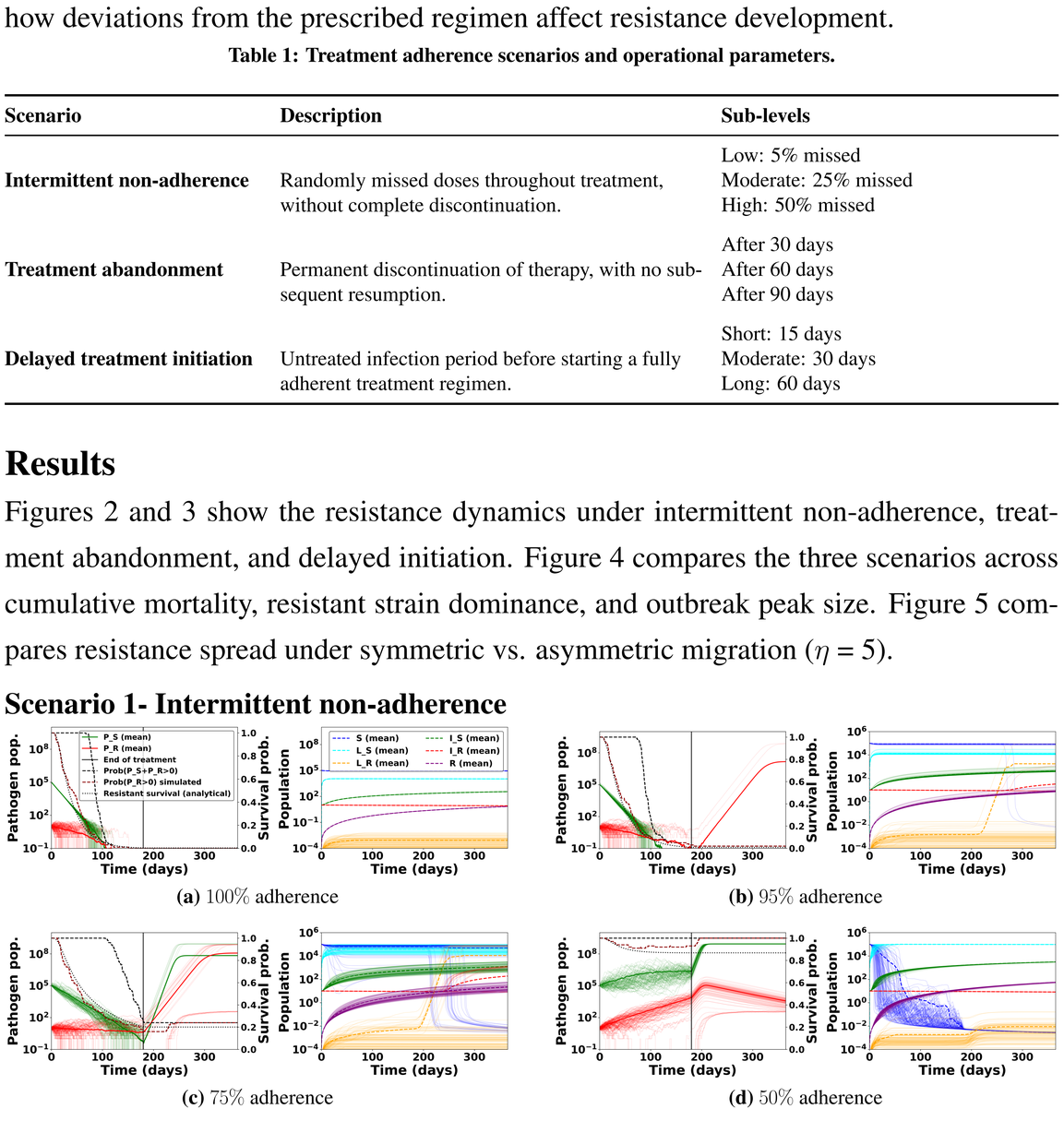
Table 1: Treatment adherence scenarios and operational parameters.

Scenario	Description	Sub-levels
Intermittent non-adherence	Randomly missed doses throughout treatment, without complete discontinuation.	Low: 5% missed Moderate: 25% missed High: 50% missed
Treatment abandonment	Permanent discontinuation of therapy, with no subsequent resumption.	After 30 days After 60 days After 90 days
Delayed treatment initiation	Untreated infection period before starting a fully adherent treatment regimen.	Short: 15 days Moderate: 30 days Long: 60 days

Results

Figures 2 and 3 show the resistance dynamics under intermittent non-adherence, treatment abandonment, and delayed initiation. Figure 4 compares the three scenarios across cumulative mortality, resistant strain dominance, and outbreak peak size. Figure 5 compares resistance spread under symmetric vs. asymmetric migration ($\eta = 5$).

Scenario 1- Intermittent non-adherence



(a) 100% adherence

(b) 95% adherence

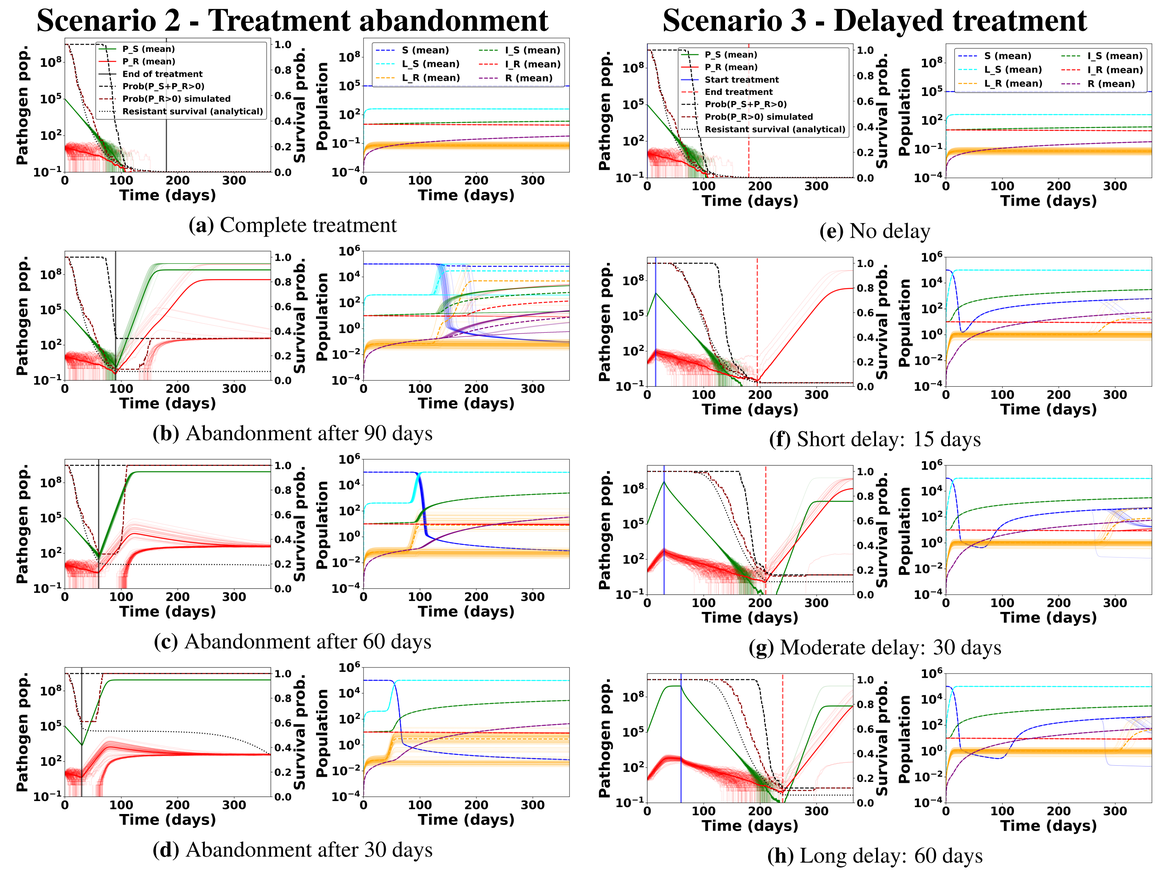
(c) 75% adherence

(d) 50% adherence

Figure 2: Adherence effects at 100%, 95%, 75%, and 50% on multiscale resistance dynamics. Panels $a - d$ correspond to the four adherence rates. Source: Author.

Scenario 2 - Treatment abandonment

Scenario 3 - Delayed treatment



(a) Complete treatment

(b) Abandonment after 90 days

(c) Abandonment after 60 days

(d) Abandonment after 30 days

(e) No delay

(f) Short delay: 15 days

(g) Moderate delay: 30 days

(h) Long delay: 60 days

Figure 3: Resistance dynamics under treatment abandonment (a–d) and delayed initiation (e–h). Pathogen population (left axis) and host subpopulations (right axis). Source: Author.

Synthesis of treatment non-adherence scenarios

Intermittent non-adherence shows the smallest effects, while treatment abandonment and delayed initiation produce the largest increases in mortality and outbreak size.

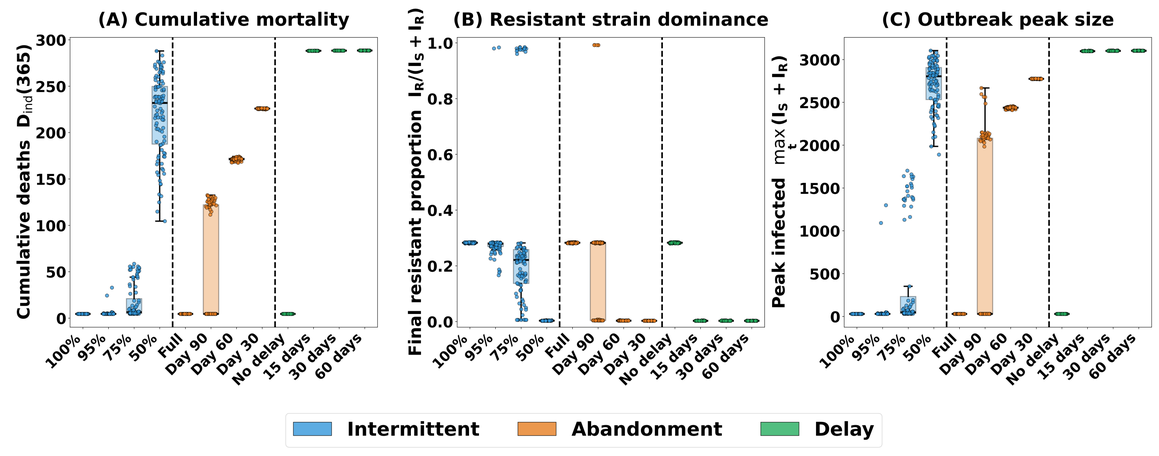
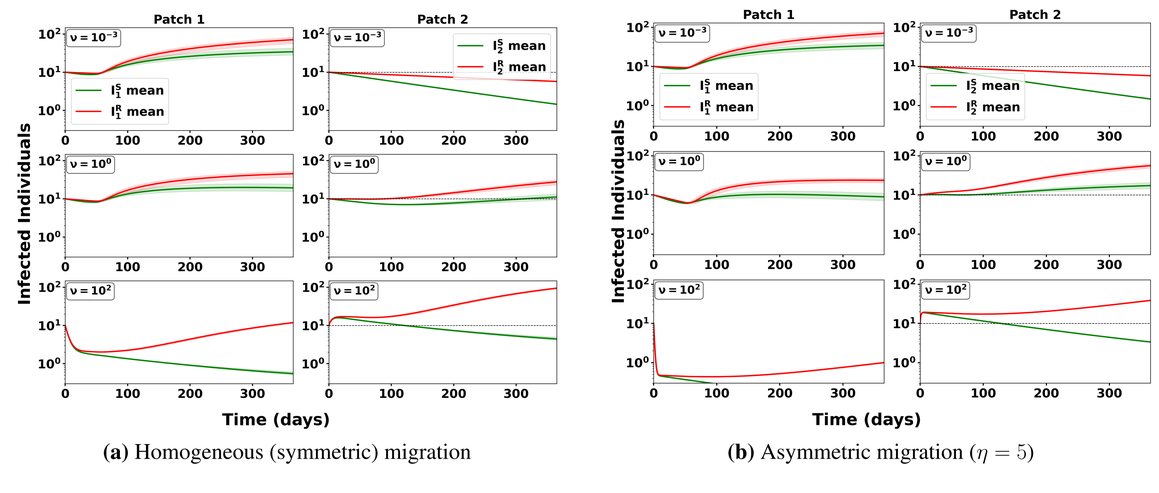


Figure 4: Comparative synthesis of treatment non-adherence scenarios. Source: Author.

Spatial spread: symmetric vs. asymmetric migration

Asymmetric mobility allows resistant lineages from non-adherent hotspots to seed outbreaks in adherent communities.



(a) Homogeneous (symmetric) migration

(b) Asymmetric migration ($\eta = 5$)

Figure 5: Migration symmetry shapes resistance spread under heterogeneous adherence: (a) homogeneous vs. (b) asymmetric migration ($\eta = 5$). Source: Author.

Conclusions

- Treatment non-adherence can substantially increase mortality and outbreak size, particularly when adherence falls to 50% or treatment is abandoned early.
- Delayed treatment initiation ($\geq$ 15 days) produces the most severe epidemiological outcomes.
- The decline in the resistant fraction (ρ_R) does not necessarily indicate a reduction in the absolute resistant burden.
- Asymmetric mobility enables the spread of resistant lineages from non-adherent hotspots, triggering delayed outbreaks in adherent communities.
- Effective control requires combining treatment adherence with targeted infection control in high-risk communities.

References

Scan the QR code to access the complete poster and reference list.